WELLSLING · FEASIBILITY ASSESSMENT

WellSling feasibility assessment

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WSL-FEA-001	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsling		
Source	docs/08-feasibility.md @ 346fa2d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade), closed for the chosen design at P3

WellSling feasibility assessment

WellSling is feasible on paper: the frame, winch, rigging and bridge work with every steel part at a factor of 3 or more and every made part can be cut, drilled and welded in a village workshop. Its feasibility now rests on four things paper cannot settle: whether a 600 kg animal tolerates the bands for about half an hour, whether the passer can be steered under a lying animal 15 m down, the soil and rim at real wells, and the handle force of the winch actually bought. Method: STANDARDS section 23.

> **Safety:** Feasibility on paper is not fitness for use. The kit is not used on an animal before a proof-load test at 1.5 times the hook load.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Statics, strength, stability and clearances all close on paper; four details that failed in kit 1.7 are fixed (leg tube, chain eyes, winch bolts, band eyes)	High	Margins are screening factors, not a certified rating; the passer and band questions remain	WSL-STM-001; WSL-CAL-001 v0.2
Materials	Common structural tube, box section, plate, hardwood and exterior plywood; galvanised	High	Tube must be structural (310 MPa), not water pipe; the mill certificate is checked	WSL-CAL-001 assumption 11; BOM lines 1, 2
Manufacturing and assembly	Every made part is cut, drilled, bent or welded from stock; every bolted and pinned joint is modelled with its holes and fasteners and checked for interference at D3	High	Galvanising needs a plant within reach; drilling lug pairs together keeps the pins true	WSL-BLD-001; `cad/fidelity.json`
Power and controls	Human power only: a crank at up to 200 N, about 94 W near the top, crew taking turns; an automatic load-pressure brake	Medium	The crank force rests on a catalogue-class handle force of 230 N at the rated load; confirm on the winch bought	WSL-CAL-001 B6; STEM S6
Supply of parts	Winch, rope, shackles, chain, slings and harnesses are catalogue items	Medium	A spur-gear winch with a load-pressure brake and a 240 mm two-ply sling with eyes for a 25 mm bar must be found locally; one cooperative buying for several villages helps	BOM lines 9, 18; MKT
Animal handling	Two plain 240 mm bands under chest and flank, about 17 to 21 kPa, about 30 minutes in the bands	Low	No cited tolerance for cattle; needs a veterinary partner before any live lift	WSL-STM-001 S25; WSL-DEC-001 item 5
Site conditions	Pads 700 mm and bearers 400 mm back from a 3 m rim; 75 kPa ground assumed	Medium	Soft, wet or undercut rims; a site check is a safety stop	WSL-STM-001 S14, S15; WSL-BLD-001
Passing the bands	A 16 m sectioned pole with a J-head and a messenger line	Low	The push the pole can give 15 m down and the force to slide under an animal in mud are not known	WSL-STM-001 S26

Ratings: High, Medium, Low.